Draft section for insertion after Section 17 (numbering is standalone here; cross-references to the paper's existing results are hard-coded by number).

18 The held rough error and the closure beyond the lazy regime

Proposition 20 says where every expansion about θ_0 stops: a rough direction is fitted only after a displacement of order the analyticity radius R, so no series in $\theta-\theta_0$ describes the stretch of training in which the rough part of the error is removed. This section changes the expansion variable. The one-time theory of Part I bounds the eigenvalue of the kernel on a rough direction by the peak feature roughness, $\lambda_k R(v_k)^2 \le \sigma_R^2$ (Corollary 1), and Proposition 15 gives the exact rate of σ_R along training. Together they say that the part of the error that a roughness function counts as rough can lose energy only through one scalar function of time, $\sigma_R(t)$, up to a leak from the smooth part; and the a priori bound on the growth of σ_R then gives a floor on the rough part of the error that holds at every time, with no ball and no horizon, and that decays at most as the exponential of a cubic in t. Read across a nested family of seminorms, the floor says that the error at time t is at least the energy of the initial error above a cutoff mode that moves up as σ_R grows. The last part of the section states the one assumption under which the same quantities give an upper bound, and the reduced two-variable model of ignition that follows from it. Everything before that assumption is exact.

Throughout, $R(u) = \|Pu\|$ is a projector seminorm, with P a symmetric projector on sampled functions, so that $C_R = 1$ and $\sigma_R(\theta)^2 = \lambda_{\max}\big(PK(\theta)P\big)$; the cluster seminorm and the manifold high-pass of Section 4.1 are of this form. Write

$$E_r = PE, \qquad E_s = (I-P)E, \qquad \sigma(t) = \sigma_R(\theta(t)), \tag{1}$$

for the rough and the smooth part of the error under R and for the peak feature roughness along the path. Since Y is fixed, $\dot E_r = P\dot E$, and $\|E_r(0)\| = R(E_0) \ge R(Y) - R(\hat Y_0)$.

18.1 The rough part loses energy only through σ_R

Proposition 1 (Energy of the rough part)**.** *Let N_θ be continuously differentiable in θ and let $\theta(t)$ be the gradient-flow path. Then, with the notation of Eq.* (1)*:*

(a) Exactly,

$$\frac{d}{dt}\,\|E_r\|^2 = -2\,\langle E_r, KE_r\rangle - 2\,\langle E_r, KE_s\rangle, \qquad 0 \le \langle E_r, KE_r\rangle \le \sigma^2\,\|E_r\|^2, \qquad |\langle E_r, KE_s\rangle| \le \sigma\,\lambda_1^{1/2}\,\|E_r\|\,\|E_s\| \tag{2}$$

(b) For every $\varepsilon > 0$ and every t,

$$\|E_r(t)\|^2 \;\ge\; \|E_r(0)\|^2\,\exp\Big(-(2+\varepsilon)\int_0^t \sigma(s)^2\,ds\Big) - \frac{1}{\varepsilon}\int_0^t \lambda_1(s)\,\|E_s(s)\|^2\,\exp\Big(-(2+\varepsilon)\int_s^t \sigma(\tau)^2\,d\tau\Big)\,ds. \tag{3}$$

(c) The rate at which the rough part can lose energy is bounded by the peak feature roughness alone once the smooth part is gone: $-\frac{d}{dt}\|E_r\|^2 \le 2\sigma^2\|E_r\|^2 + 2\sigma\lambda_1^{1/2}\|E_r\|\,\|E_s\|$.

Proof. (a) $\frac{d}{dt}\frac12\|E_r\|^2 = \langle PE, P\dot E\rangle = \langle E_r, \dot E\rangle = -\langle E_r, KE\rangle$ by Theorem 7(a), and $E = E_r + E_s$. K is positive semidefinite, so $\langle E_r, KE_r\rangle \ge 0$. For a unit ψ in the range of P, $R(\psi) = \|\psi\| = 1$, so $\psi \in R^\circ$

and $\|\psi\|_K \le \sigma_R$ by the definition of σ_R (proof of Theorem 1(a)); hence $\langle E_r, KE_r\rangle = \|E_r\|_K^2 \le \sigma^2\|E_r\|^2$. For the cross term, $\langle a, Kb\rangle = g_a \cdot g_b$, so $|\langle E_r, KE_s\rangle| \le \|E_r\|_K\|E_s\|_K \le \sigma\|E_r\| \cdot \lambda_1^{1/2}\|E_s\|$. (b) By (a) and $2ab \le \varepsilon a^2 + b^2/\varepsilon$ with $a = \sigma\|E_r\|$ and $b = \lambda_1^{1/2}\|E_s\|$,

$$\frac{d}{dt}\|E_r\|^2 \;\ge\; -(2+\varepsilon)\,\sigma^2\,\|E_r\|^2 \;-\; \frac{\lambda_1}{\varepsilon}\,\|E_s\|^2,$$

a linear differential inequality for the nonnegative function $\|E_r\|^2$; Grönwall's inequality in its integrating-factor form gives Eq. (3). (c) is (a) with the two bounds. □

Part (a) is the energy identity of Theorem 7(a) projected on the rough subspace. Its content is that the kernel enters the energy of the rough part only through two numbers at each instant: its Rayleigh quotient on the rough subspace, capped by σ^2, and its coupling of the rough subspace to the smooth one, capped by $\sigma\lambda_1^{1/2}$. Part (b) is the dynamic form of the floor of Theorem 4. Where Theorem 8(b) charges the path linearly, $\int \sigma_R|\dot\theta|$, Eq. (3) charges it exponentially, $\exp(-(2+\varepsilon)\int\sigma^2)$, and the two are different readings of the same fact, that the output can acquire roughness only at rate σ_R. The second term is the leak: the smooth part of the error drives the rough part through the off-diagonal block of the kernel, and it is small once the smooth part has been removed, which the floor of Part I says happens while the rough part stays. With $\varepsilon \to 0$ the exponent tends to the rate $2\sigma^2$ of Corollary 1 and the leak coefficient grows; for a bound that is to be read along a run, ε is chosen to balance the two terms. Along a run, $\sigma(t)$ is a top eigenvalue of PKP at each saved state and $\|E_s(t)\|$ is read off, so Eq. (3) is a check with no free constant.

18.2 The a priori floor on the rough part

Proposition 15(b) bounds the growth of σ_R along gradient flow by $\Theta_R \le \sigma_R\lambda_1^{1/2}H_2\|E\|$ (with $C_R = 1$). Inserted into Proposition 1, this gives a floor on the rough part that is computed from the initial state and holds at every time.

Theorem 2 (The rough part is held: a priori form)**.** *Let N_θ be twice continuously differentiable in θ, let $\theta(t)$ be the gradient-flow path, and let $\bar\lambda_1$ and $\bar H_2$ be the largest values of $\lambda_1(\theta(s))$ and $H_2(\theta(s))$ for $s \le t$ (or over the ball of radius $\ell(t)$ about θ_0, which contains the path). Let $\sigma_0 = \sigma_R(\theta_0)$ and*

$$a = \bar\lambda_1^{1/2}\,\bar H_2\,\|E_0\|, \qquad \Sigma(s) = \sigma_0 + a\,s, \qquad I(t) = \int_0^t \Sigma(s)^2\,ds = \sigma_0^2\,t + \sigma_0\,a\,t^2 + \tfrac{1}{3}\,a^2\,t^3. \tag{4}$$

Then $\sigma(s) \le \Sigma(s)$ for all $s \le t$, and for every $\varepsilon > 0$,

$$\|E_r(t)\|^2 \;\ge\; \|E_r(0)\|^2\,e^{-(2+\varepsilon)I(t)} \;-\; \frac{\bar\lambda_1}{\varepsilon}\,J_s(t), \qquad J_s(t) = \int_0^t \|E_s(s)\|^2\,ds. \tag{5}$$

In particular $\|E(t)\| \ge \|E_r(t)\|$ obeys the same bound, and $\|E_r(0)\| \ge R(Y) - R(\hat Y_0)$.

Proof. Let $S(t) = \sigma_0^2 + 2\int_0^t \Theta_R$, so that $\sigma(t)^2 \le S(t)$ by Eq. (98) and $S' = 2\Theta_R \le 2\sigma\lambda_1^{1/2}H_2\|E\| \le 2S^{1/2}\bar\lambda_1^{1/2}\bar H_2\|E_0\|$, using $\sigma \le S^{1/2}$ and $\|E\| \le \|E_0\|$ (Theorem 7(a)). Hence $(S^{1/2})' \le a$ and $\sigma(s) \le S(s)^{1/2} \le \sigma_0 + as = \Sigma(s)$. Insert $\sigma \le \Sigma$ into Eq. (3), bound $\lambda_1(s)$ by $\bar\lambda_1$ and the inner exponential in the leak term by one. □

The theorem is the counterpart, for the energy of the rough part, of the minimum-path-length statement of Corollary 5. Three features distinguish it. It holds at every time with no horizon and

no analyticity, since nothing is expanded. Its decay is at most exponential in a cubic: with no rough features at the start, $\sigma_0 = 0$, which is the situation of Proposition 20 for a network with a smooth activation, the exponent is $\frac{2+\varepsilon}{3}a^2t^3$, so the rough part is held for a time of order $a^{-2/3}$. And the kernel enters only through σ_0, $\bar{\lambda}_1$ and $\bar{H}_2$: the initial peak feature roughness, the top eigenvalue, and the second-derivative size, which are the three numbers through which an architecture enters the rest of this section. The quantity $J_s(t)$ is the time integral of the energy of the smooth part; a priori it is bounded by the band of Theorem 10 about $e^{-K_0 t}E_0$ projected on $I - P$, and along a run it is read off.

Corollary 3 (Minimum time to remove the rough part)**.** *Under the hypotheses of Theorem 2, if* $\|E_r(t)\| \le \epsilon$ *with* $\epsilon^2 + \bar{\lambda}_1 J_s(t)/\varepsilon < \|E_r(0)\|^2$*, then*

$$(2+\varepsilon)\, I(t) \;\ge\; \log \frac{\|E_r(0)\|^2}{\epsilon^2 + \bar{\lambda}_1 J_s(t)/\varepsilon}, \tag{6}$$

and in particular, when $\sigma_0 = 0$*,*

$$t \;\ge\; \Big(\frac{3}{(2+\varepsilon)\, a^2}\, \log \frac{\|E_r(0)\|^2}{\epsilon^2 + \bar{\lambda}_1 J_s(t)/\varepsilon}\Big)^{1/3}. \tag{7}$$

Proof. Rearrange Eq. (5); I is increasing in t, and for $\sigma_0 = 0$, $I(t) = a^2t^3/3$. □

Eq. (7) is the a priori ignition time of a rough direction from a smooth start, and it depends on the architecture through $a = \bar{\lambda}_1^{1/2}\bar{H}_2\|E_0\|$ alone: the fastest the peak feature roughness can grow per unit time is $\bar{\lambda}_1^{1/2}\bar{H}_2$ per unit of error, and the rough part is held until the integrated square of that growth is of order one. Like Corollary 5 it is one-sided; unlike it, the time scale is a cube root rather than a square, because the roughness supply itself grows linearly while the error is held.

18.3 The roughness spectrum and the moving cutoff

The floor of Theorem 2 holds for every projector seminorm. Read across a nested family, it locates where the held error sits. Let $P_1 \succeq P_2 \succeq \cdots$ be a nested family of projectors, each the projector of a seminorm $R_m(u) = \|P_m u\|$; for the manifold high-pass of Section 4.1, P_m removes the first m graph-Laplacian modes, and the range of P_m shrinks as m grows. Write $\sigma_m(\theta)^2 = \lambda_{\max}(P_m K P_m)$, which is nonincreasing in m, and

$$\rho_m = \|P_m E_0\|, \qquad I_m(t) = \sigma_m(\theta_0)^2\, t + \sigma_m(\theta_0)\, a\, t^2 + \tfrac{1}{3}a^2t^3. \tag{8}$$

$m \mapsto \rho_m$ is the roughness spectrum of the initial error: the energy of E_0 above mode m. It is fixed by the dataset and the initial output, and for a target with $\hat{Y}_0$ small it is the roughness spectrum of the labels.

Definition 4 (Cutoff mode)**.** For $t \ge 0$ and $\varepsilon > 0$, the cutoff mode is

$$m^\star(t) = \min\{\, m : (2+\varepsilon)\, I_m(t) \le \log 2 \,\}, \tag{9}$$

the lowest mode above which, at time t, the a priori bound allows the rough part to have lost at most half its energy.

Since I_m is nonincreasing in m and increasing in t, the set in Eq. (9) is an up-set in m and $m^\star(t)$ is nondecreasing in t: the cutoff moves up through the modes as training proceeds, and it moves even when $\sigma_m(\theta_0) = 0$, through the a^2t^3 term, which is feature learning.

Proposition 5 (The error at time t is at least the energy above the cutoff)**.** *Under the hypotheses of Theorem 2, for every t,*

$$\|E(t)\|^2 \;\geq\; \|P_{m^\star(t)}E(t)\|^2 \;\geq\; \tfrac{1}{2}\,\rho^2_{m^\star(t)} \;-\; \frac{\bar\lambda_1}{\varepsilon}\int_0^t \|(I-P_{m^\star(t)})E(s)\|^2\,ds. \tag{10}$$

Proof. Theorem 2 with $P = P_{m^\star(t)}$ and $e^{-(2+\varepsilon)I_{m^\star}(t)} \geq \frac{1}{2}$ by the definition of $m^\star$. □

In words: at time t, the error is at least the part of the initial error that is rougher than the cutoff mode, up to the leak, and the cutoff is set by the roughness capacity of the network, $\sigma_m(\theta_0)$ and a, and by nothing else. Without feature learning ($a = 0$) the cutoff is the smallest m with $\sigma_m(\theta_0)^2 t \leq \log 2/(2+\varepsilon)$, which is the spectral bias of the lazy regime stated in roughness rather than in frequency; with feature learning the cutoff rises faster, and Corollary 3 says how much faster it can rise at most. This is the quantity that the closure below takes as its estimate of the held rough error: for a dataset whose roughness spectrum has a step at some mode m_Y, the error is held near ρ_{m_Y} until $m^\star(t)$ passes m_Y, and the time at which it does is Eq. (6) for $P = P_{m_Y}$.

18.4 The closure: efficient roughness supply

The statements above bound the rough part from below. An upper bound, and with it a prediction, needs one hypothesis, and it is the one that Section 14 identified as the closure that none of its assumptions provides: the growth of the kernel on the rough directions with the sign that learning needs. The following is its quantitative form.

Assumption 6 (Efficient roughness supply)**.** There is $c \in (0,1]$ such that, along the stretch $[t_0, t]$ of gradient flow,

$$\langle E_r, KE_r\rangle \;\geq\; c\,\sigma_R^2\,\|E_r\|^2, \tag{11}$$

that is, the Rayleigh quotient of the kernel on the current rough residual is at least the fraction c of its largest value on the rough subspace.

The assumption says that the roughness that the features carry is carried along the residual rather than along rough directions that carry no error. It is the positive, quantitative form of the failure of Assumption 4 on the rough directions, and it has a mechanism: by Proposition 16, $\dot K$ is supported on the second-order features $\nabla^2_\theta N_\theta[g_E]$ of the current gradient, and once the smooth part has been removed, $g_E \approx g_{E_r}$, so the kernel grows along the second-order companions of the rough residual itself. Whether it grows along the residual's own direction, with c of order one, is the measurement: one kernel product $\langle E_r, KE_r\rangle$ and one top eigenvalue of PKP at each saved state. The assumption is a statement about the direction of the roughness supply, not about its amount, which remains $\sigma_R(t)$.

Proposition 7 (Two-sided band for the rough part)**.** *Under the hypotheses of Proposition 1 and Assumption 6 on $[0,t]$, for every $\varepsilon \in (0, 2c)$,*

$$\|E_r(t)\|^2 \;\leq\; \|E_r(0)\|^2\,e^{-(2c-\varepsilon)\int_0^t \sigma^2} \;+\; \frac{1}{\varepsilon}\int_0^t \lambda_1(s)\,\|E_s(s)\|^2\,e^{-(2c-\varepsilon)\int_s^t \sigma^2}\,ds, \tag{12}$$

which with Eq. (3) *encloses $\|E_r(t)\|^2$ in a band in which the kernel enters only through $\sigma(t)$ and the leak.*

Proof. From Eq. (2) and Eq. (11), $\frac{d}{dt}\|E_r\|^2 \leq -2c\sigma^2\|E_r\|^2 + 2\sigma\lambda_1^{1/2}\|E_r\|\|E_s\| \leq -(2c-\varepsilon)\sigma^2\|E_r\|^2 + (\lambda_1/\varepsilon)\|E_s\|^2$, and Grönwall's inequality. □

Under the assumption, the rough part of the error is predicted by one scalar function of time, $\sigma_R(t)$, to within the band set by c and the leak. What remains open is $\sigma_R(t)$ itself: Proposition 15(b) gives its rate exactly, $\frac{d}{dt}\sigma_R^2 \le 2\Theta_R$ with Θ_R the largest drive over the dual ball, and bounds it by $\sigma_R\lambda_1^{1/2}H_2\|E\|$; its actual value needs the second derivative of the network along the path, which is where the frozen-residual reduced flow of the last part enters. The following reduced model is what the band becomes when both the supply and its growth are taken at their bounds.

Proposition 8 (Reduced model of ignition)**.** *Let $r = \|E_r\|$ and suppose that on a stretch of training the smooth part has been removed, $E \approx E_r$, that Assumption 6 holds with constant c, and that the drive attains its bound with the favourable sign, $\Theta_R = \sigma_R\lambda_1^{1/2}H_2\|E\|$, with λ_1 and H_2 treated as constants $b = \lambda_1^{1/2}H_2$. Then (r, σ) obey*

$$\dot r = -c\,\sigma^2\,r, \qquad \dot\sigma = b\,r, \tag{13}$$

which have the first integral $r + \frac{c}{3b}\sigma^3 = \text{const}$. Hence the peak feature roughness that the network reaches when the rough residual r_0 has been removed is

$$\sigma_\infty = \Big(\sigma_0^3 + \frac{3b}{c}\,r_0\Big)^{1/3}, \tag{14}$$

and the time at which r has fallen to $r_0/2$ is $t_{1/2} = \int_{\sigma_0}^{\sigma_{1/2}} d\sigma \big/ \big(b\,r_0 - \frac{c}{3}(\sigma^3 - \sigma_0^3)\big)$ with $\sigma_{1/2}^3 = \sigma_0^3 + 3br_0/(2c)$; for $\sigma_0 = 0$ this is $t_{1/2} = \kappa_c\,(b^2 r_0^2 c)^{-1/3}$ with $\kappa_c = \int_0^{(3/2)^{1/3}} du/(1-u^3/3)$, a number.

Proof. The first equation is Proposition 1(a) with the cross term dropped and the quadratic form at its assumed value; the second is $\frac{d}{dt}\sigma^2 = 2\Theta_R = 2\sigma b r$. Then $dr/d\sigma = -c\sigma^2/b$, which integrates to the first integral; set $r = 0$ for σ_∞, and $dt = d\sigma/(br)$ with r from the first integral for $t_{1/2}$; the substitution $\sigma = (3br_0/c)^{1/3}u$ gives the stated form. □

Eq. (14) is a statement of what the kernel needs: to fit a rough residual of size r_0, the features must become rough to the level σ_∞, and the network supplies roughness at rate at most b per unit of error. It is the counterpart in roughness of Proposition 20 in displacement, and unlike that proposition it is a model rather than a bound: its two inputs, c and the realized Θ_R against its bound, are measured. The cube-root scaling of $t_{1/2}$ with $b^2r_0^2$ is the one of Corollary 3, which holds without the model; the model adds the constant and the second half of the curve, the decay of r once σ has grown, which is exponential at the rate $c\sigma_\infty^2$ as $r \to 0$.

18.5 Depth and the comparison of architectures

Everything above enters an architecture through three numbers at the start of training, $\sigma_R(\theta_0)$, $\bar\lambda_1$ and $\bar H_2$, and through the two run quantities c and Θ_R. For two designs on the same dataset, the comparison principle of Corollary 6 applies with the bounds of this section.

Corollary 9 (Comparing designs on the rough part)**.** *Let designs A and B be trained by gradient flow on the same dataset, let P be the projector of a roughness function fixed by the inputs, and let $T > 0$.*

(i) (Proved.) $\|E_r^A(T)\|^2$ is at least the right side of Eq. (5) computed with σ_0^A, a_A and $J_s^A(T)$.

(ii) (Under Assumption 6 for B with constant c_B and with $\sigma_B(t)$ read off the run, or taken from the reduced model.) $\|E_r^B(T)\|^2$ is at most the right side of Eq. (12).

(iii) If the bound in (ii) is below the bound in (i), then $\|E_r^B(T)\| < \|E_r^A(T)\|$: B has learned more of the rough part of the labels by time T.

The corollary separates what depth does into two effects that can be checked one at a time. A deeper network can have larger initial peak feature roughness $\sigma_R(\theta_0)$, because its gradient features decorrelate over shorter input distances when the initialization is near the edge of chaos; that lowers the held rough error after the lazy phase, since the cutoff of Definition 4 starts higher. And it can have larger $\lambda_1^{1/2} H_2$, which is the rate b at which it can supply roughness per unit of error; that shortens the ignition time of Corollary 3 and Proposition 8. Both are computed before training, one top eigenvalue of PKP and one nested automatic differentiation per design, and both are expected to grow with depth; the theory says which part of the error each one removes and in what time, but not that either grows with depth, which is a measurement or a statement about the initialization. Where the initialization statistics of deep networks are known in closed form, they can supply $\sigma_R(\theta_0; L)$, $\lambda_1(L)$ and $H_2(L)$ as functions of depth, and this is the only place in the theory where a limit of the initialization enters; nothing along the path is taken to a limit.

18.6 Scope and what to measure

Exact under gradient flow, no hypothesis beyond smoothness. Proposition 1; Theorem 2 and Corollary 3 with $\bar{\lambda}_1$ and $\bar{H}_2$ taken along the path or over the ball of radius $\ell(t)$; Proposition 5; Corollary 9(i). The leak term is exact and is read off a run; its a priori bound through Theorem 10 is available while the band of that theorem is informative. **Under Assumption 6.** Proposition 7; Corollary 9(ii)–(iii). **Model.** Proposition 8, which takes the supply and its growth at their bounds. The present section breaks the rule of Part III that nothing is approximated, in its last two parts only, and it does so with a proved band around the approximation.

The measurements that decide whether the closure is the right one, to be added to the list of Section 15: (13) $\sigma_R(t)$ along each run for the manifold high-pass at several m, against its a priori bound $\Sigma(t)$ of Eq. (4) and against the realized drive Θ_R; (14) the ratio $\langle E_r, KE_r\rangle/(\sigma_R^2\|E_r\|^2)$ along each run, which is the constant c of Assumption 6 and the quantity that decides whether the roughness supply is spent on the residual; (15) $\|E_r(t)\|^2$ against the two sides of the band, Eqs. (3) and (12), and $J_s(t)$; (16) the cutoff mode $m^\star(t)$ against the mode at which the measured error's roughness spectrum actually drops, which tests Proposition 5 as an estimate rather than as a bound; (17) $\sigma_R(\theta_0)$, λ_1 and H_2 as functions of depth at initialization, which are the three numbers through which depth enters Corollary 9.